

CAB BOOKING(Cab-U)

Full-Stack Cab Booking Management System (MERN)

Date: 6 July 2026

1. Introduction

Project Title: CAB BOOKING (FULL – STACK MERN)

Project Nature: A comprehensive full-stack cab booking platform built using the MERN stack. It connects passengers with drivers in real-time, providing a seamless interface for ride requests, fare estimation, and fleet management.

Team Members:

THANDRA MANIKANTA – Full Stack Developer (Frontend, Backend, Database, Testing & Documentation).

2. Project Overview

Purpose: To provide a reliable and user-friendly digital cab booking platform that connects passengers with nearby drivers. The system manages the complete ride lifecycle, including user registration, ride booking, driver assignment, trip tracking, fare calculation, payment, and ride history.

Key Features:

- User Registration and Secure Login (Passenger, Driver, Admin).
- Real-time Ride Booking with Nearby Driver Assignment.
- Dynamic Fare Calculation based on distance and vehicle type.
- Live GPS Tracking and Route Navigation using Maps.
- Ride History, Booking Details, and Digital Trip Records.
- Multiple Payment Options (Cash, UPI, Cards - Future Enhancement).
- Admin Dashboard for managing users, drivers, and bookings.
- Driver Profile and Availability Management.
- Ride Status Updates (Requested, Accepted, Ongoing, Completed, Cancelled).
- Responsive Web Interface built using the MERN Stack.

3. Architecture

Frontend (React.js)

Built using React 18 for the frontend and Node.js with Express.js for the backend. MongoDB is used for data storage, and the application provides user authentication, cab booking, and ride management through a responsive web interface..

Components include:

- Home Page
- Login
- Signup
- Book Ride
- Ride History
- User Dashboard
- Admin Dashboard

Backend (Node.js & Express.js)

Backend is developed using

- Node.js
- Express.js

Backend provides REST APIs for

- Authentication
- Ride Booking
- User Management
- Driver Management
- Payments

Security includes

- JWT Authentication
- bcrypt Password Encryption

Database (MongoDB)

Backend is developed using

- Node.js
- Express.js

Backend provides REST APIs for

- Authentication
- Ride Booking
- User Management
- Driver Management
- Payments

Security includes

- JWT Authentication
- bcrypt Password Encryption

4. Setup Instructions

Prerequisites

- Node.js
- MongoDB Atlas
- VS Code
- Git
- npm

Installation

Clone Project

```
git clone https://github.com/username/ride-ready.git
```

Install Frontend

```
cd client  
npm install
```

Install Backend

```
cd server  
npm install
```

Environment Variables

Create .env

```
MONGODB_URI=mongodb://localhost:27017/tmk
```

```
JWT_SECRET=secret@2025
```

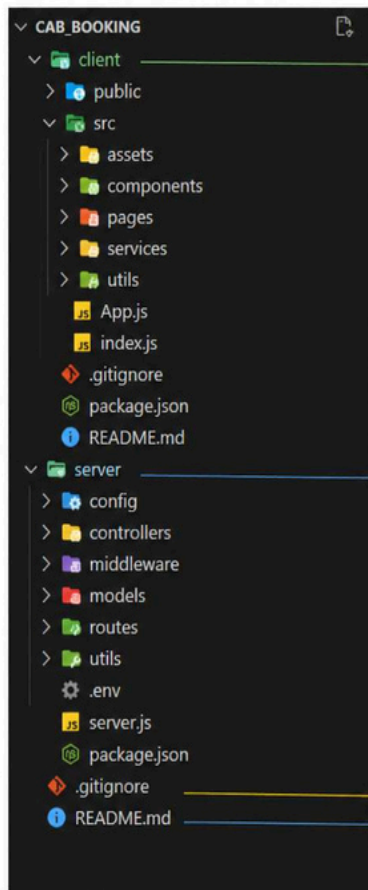
```
IMAGEKIT_PUBLIC_KEY=public_ctbf8v7sVuL67VLQHI9bRhOJo78=
```

```
IMAGEKIT_PRIVATE_KEY=private_OJQw95kQz68TbzhDgz247lYXc6w=
```

```
IMAGEKIT_URL_ENDPOINT=https://ik.imagekit.io/scjefmk1c
```

5. Folder Structure

Folder Structure



client/ - Frontend of the application built with React

- **public/** - Static files and HTML template
- **src/** - Source code
 - **assets/** - Images, icons, and other static assets
 - **components/** - Reusable UI components
 - **pages/** - Page components (Home, Login, Dashboard, etc.)
 - **services/** - API calls and axios configurations
 - **utils/** - Helper functions and utilities
- **App.js** - Main App component
- **index.js** - Entry point of React application

server/ - Backend of the application built with Node.js and Express

- **config/** - Database connection and configuration files
- **controllers/** - Route controllers (business logic)
- **middleware/** - Custom middleware (auth, error handling, etc.)
- **models/** - Mongoose models (User, Ride, etc.)
- **routes/** - API routes (auth, rides, users, etc.)
- **utils/** - Helper functions and utilities
- **.env** - Environment variables
- **server.js** - Entry point of the backend server
- **package.json** - Backend dependencies and scripts

.gitignore - Files and folders to be ignored by Git

README.md - Project documentation

6. Running the Application

To run the Cab Booking (Cab-U) application locally, start both the backend and frontend servers in separate terminal windows.

Backend Server

1. Open a terminal.
2. Navigate to the server directory:
`cd server`
3. Install dependencies (first time only):
`npm install`
4. Start the backend server:
`npm start`
5. The backend server will run at:
`http://localhost:5000`

Frontend Server

1. Open a new terminal.
2. Navigate to the client directory:
`cd client`
3. Install dependencies (first time only):
`npm install`
4. Start the frontend application:
`npm start`
5. The frontend application will run at:
`http://localhost:3000`

7. API Documentation

The Cab Booking (CAB-U) system uses RESTful APIs to enable communication between the React frontend and the Node.js/Express backend. These APIs handle user authentication, ride booking, ride management, driver operations, and data retrieval securely and efficiently.

Method	Endpoint	Description
POST	/api/auth/register	Register a new user account.
POST	/api/auth/login	Authenticate user and generate login session.
POST	/api/rides/book	Book a new cab ride.
GET	/api/rides/history	Retrieve the user's ride history.
GET	/api/drivers	Fetch available drivers for booking.
PUT	/api/rides/:id/status	Update ride status (Accepted, Ongoing, Completed).
DELETE	/api/rides/:id/cancel	Cancel a booked ride.
GET	/api/users/profile	Retrieve logged-in user profile information.
PUT	/api/users/profile	Update user profile details.

8. Authentication & Role-Based Security

Protected by JWT. Differentiation between 'Passenger' and 'Driver' is handled at the middleware level. Payments are secured using Stripe integration or secure Wallet systems.

Authentication uses
JWTToken
bcrypt

Workflow:-

1 User registers



Password encrypted



Saved into MongoDB



User Login



JWTToken generated



Token sent to frontend

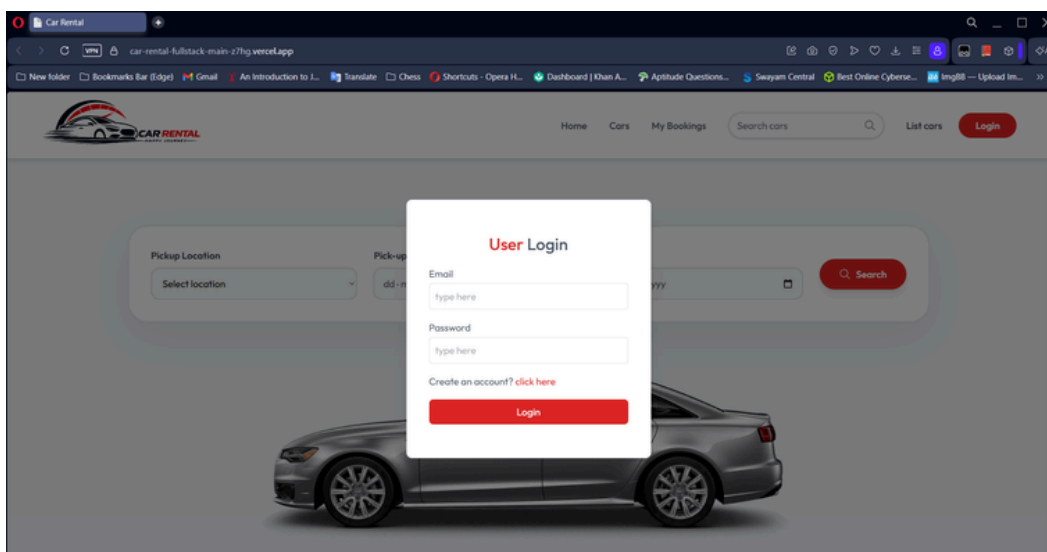


Protected APIs accessed using JWT

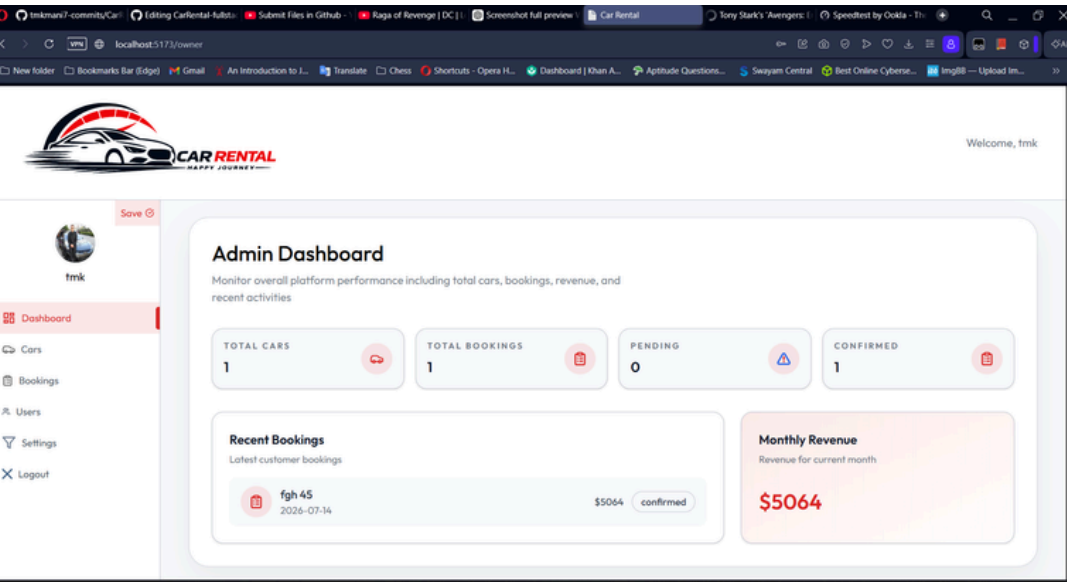
9. User Interface

The Ride Ready user interface is designed using React.js with a clean, responsive, and user-friendly layout. It provides separate screens for user registration, login, ride booking, ride tracking, payment, and admin management, ensuring a smooth device.

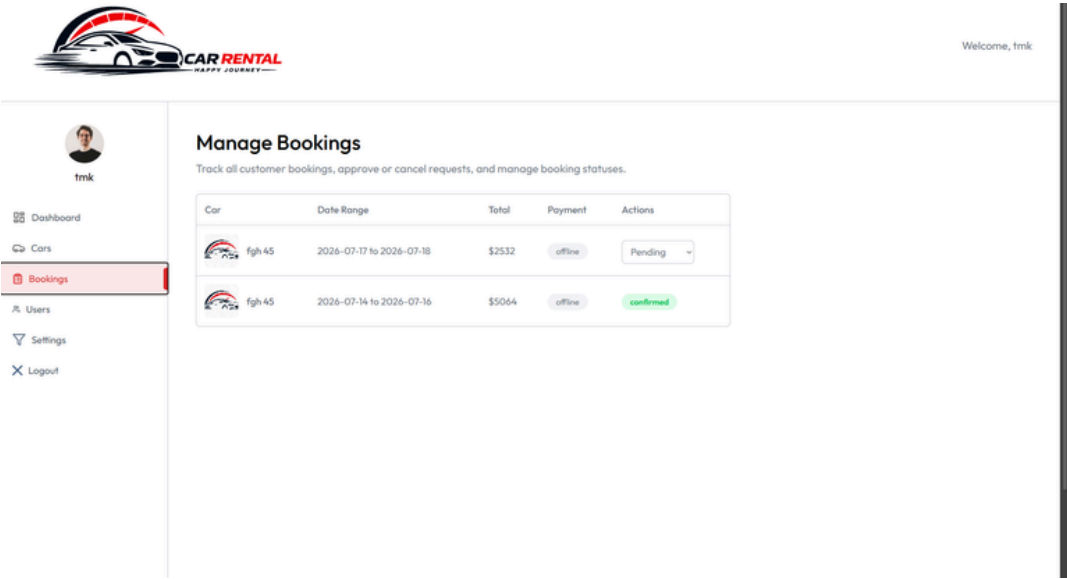
LOGIN:-



Dashboard :-



RIDE BOOKING:-



RIDE TRACKING :-

Welcome, tmk



tmk

- Dashboard
- Cars
- Bookings**
- Users
- Settings
- Logout

Manage Bookings

Track all customer bookings, approve or cancel requests, and manage booking statuses.

Car	Date Range	Total	Payment	Actions
 fgh 45	2026-07-17 to 2026-07-18	\$2532	offline	Pending
 fgh 45	2026-07-14 to 2026-07-16	\$5064	offline	confirmed

REVIEW & PAYMENT OF THE DRIVE:

Home Cars My BookingsDashboard

What Our Customers Say

See why travelers across India trust us for a premium and reliable travel experience.

**Vikram Reddy**
Hyderabad, India

★★★★★

"Booking was quick and easy. The car was delivered on time and was in excellent condition. I would definitely use this service again."

**Ananya Gupta**
Pune, India

★★★★★

"The entire process was smooth from start to finish. Customer support was very responsive and helpful."

**Karthik Rao**
Visakhapatnam, India

★★★★★

"Clean vehicles, affordable pricing, and professional service. One of the best car rental experiences I've had."

**Sneha Nair**
Kochi, India

★★★★★

"The car was well maintained and fuel efficient. The booking system was simple and

**Arjun Mehta**
Jaipur, India

★★★★★

"Excellent service and transparent pricing. The vehicle exceeded my expectations and

10. Testing & Validation

Testing includes

- User Registration Testing
- Login Testing
- Ride Booking Testing
- Payment Testing
- Driver Assignment Testing
- Authentication Testing

Testing Type

- Functional Testing
- Manual Testing
- API Testing

Expected Result

All modules work successfully.

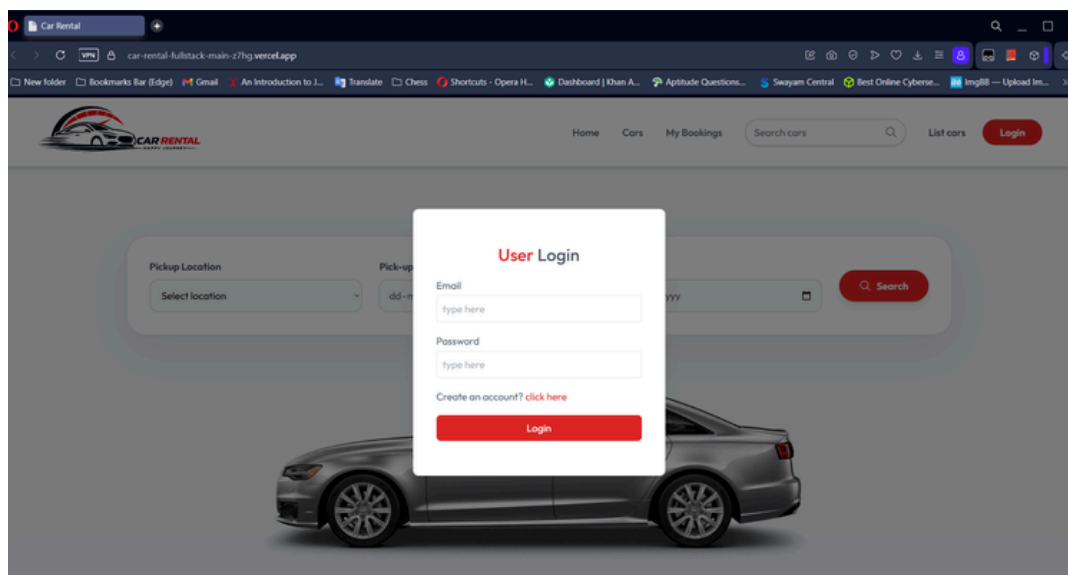
11. Screenshots or Demo

Demo Link:-

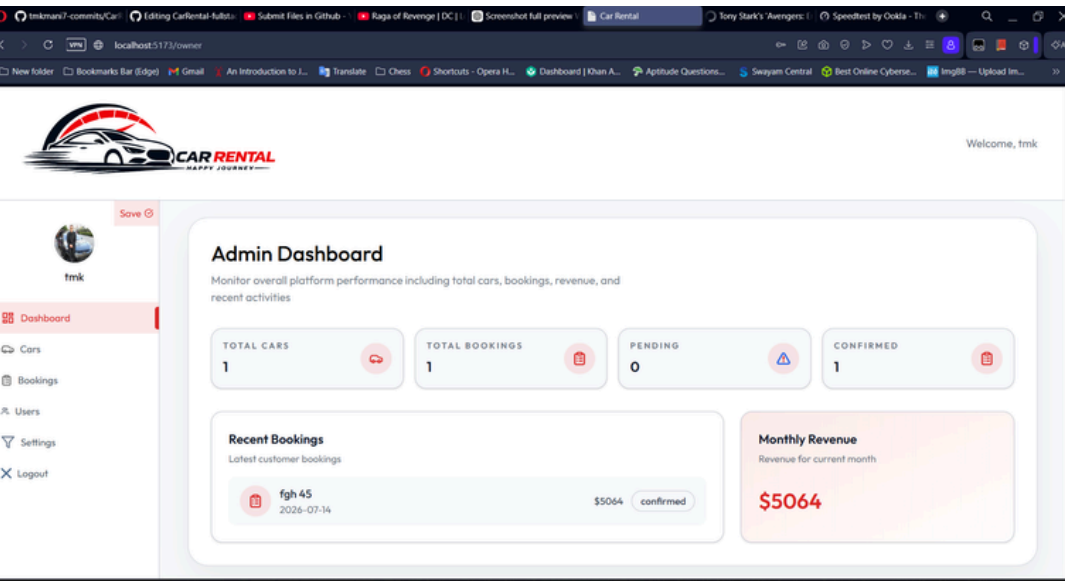
<https://github.com/tmkmani7-commits/CarRental-fullstack-main/tree/main>

Screenshots:-

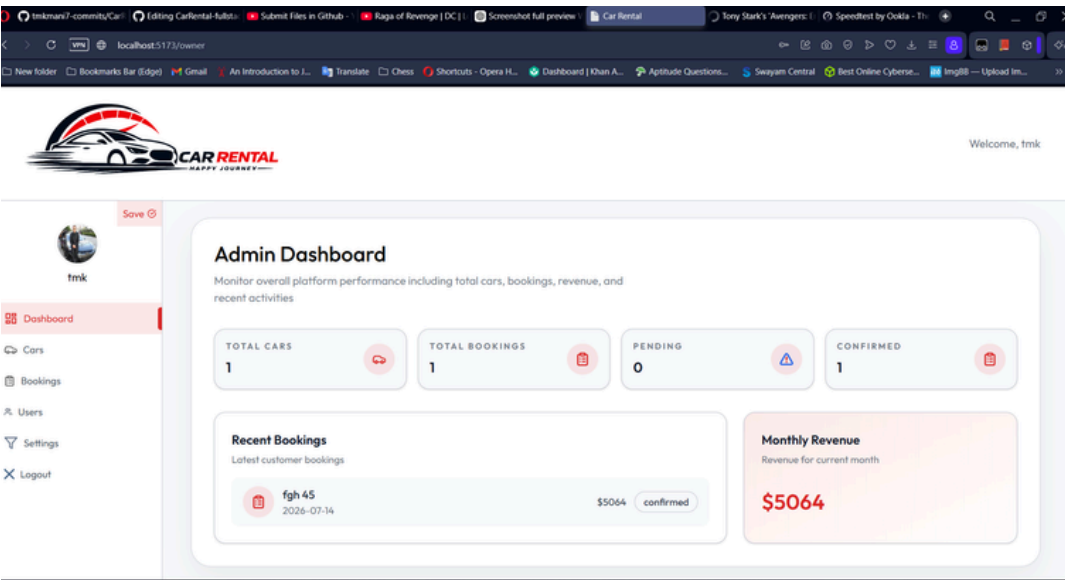
LOGIN:-



CAB BOOKING Dashboard :-



RIDE BOOKING:-



RIDE TRACKING :-

Welcome, tmk



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RIDE HISTORY:-

Welcome, tmk



tmk

- Dashboard**
- Cars
- Bookings
- Users
- Settings
- Logout

Admin Dashboard

Monitor overall platform performance including total cars, bookings, revenue, and recent activities

TOTAL CARS
1

TOTAL BOOKINGS
1

PENDING
0

CONFIRMED
1

Recent Bookings

Latest customer bookings

 fgh 45
2026-07-14

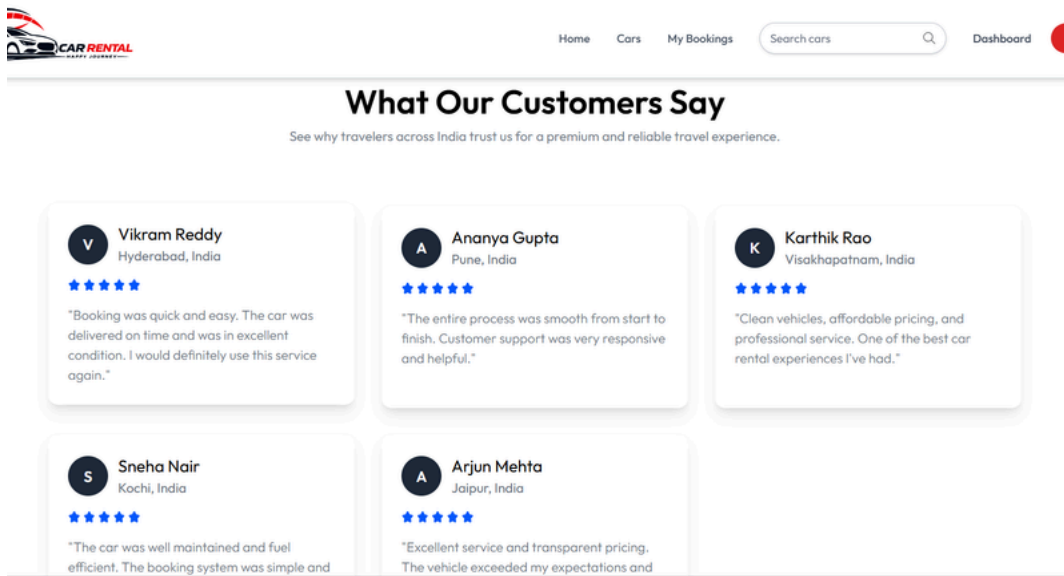
\$5064 confirmed

Monthly Revenue

Revenue for current month

\$5064

REVIEW & PAYMENT OF THE DRIVE:-



12. Known Issues

- GPS accuracy may reduce in areas with poor internet or weak network coverage.
- Live GPS tracking is currently limited and does not provide continuous real-time updates.
- Online payment gateway integration is under development and not fully functional.
- Driver availability status may experience slight delays due to network latency.
- Push notifications for ride booking and ride status updates are not yet implemented.
- The application currently supports only web browsers and does not include a dedicated mobile app.
- Ride scheduling for future bookings and advanced fare calculation features are planned for future releases.

13. Future Enhancements

Future Enhancements

- Implement real-time GPS tracking to provide accurate live driver and ride location updates.
- Integrate secure online payment gateways such as Razorpay, Stripe, and UPI for seamless digital transactions.
- Add push notifications to keep users informed about ride confirmations, driver arrivals, and payment status.
- Introduce ride scheduling to allow users to book cabs in advance for future travel.
- Develop a driver rating and customer feedback system to improve service quality and user satisfaction.
- Integrate AI-based fare estimation and smart driver allocation for faster ride matching and optimized pricing.
- Launch Android and iOS mobile applications with multilingual support and SOS emergency features for enhanced accessibility and safety.